EXPLORING THE SOIL-ASSOCIATED MICROBIOME OF COLOMBIAN COFFEE PLANTATIONS: A METABARCODING APPROACH TO BACTERIAL AND FUNGAL TAXONOMIC AND FUNCTIONAL ANALYSIS

CRISTANCHO MARCO[1], FAGGIOLI VALERIA[2], GOMEZ-GODINEZ LORENA[3], OCHOA VICTOR HUGO[4], RIVAROLA MAXIMO[2]

1.- CENICAFÉ-COLOMBIA, MARCO.CRISTANCHO@CAFEDECOLOMBIA.COM; 2.- INTA-ARGENTINA, FAGGIOLI.VALERIA@INTA.GOB.AR; 3.- NATIONAL AUTONOMOUS UNIVERSITY OF MEXICO; 4.- ARIZONA STATE UNIVERSITY

INTRODUCTION

Coffee is a major tropical crop, with Colombia as the third largest world producer. Soil microbiota from coffee producing areas is poorly described.

This study aims to describe the soil bacteria and fungi associated with coffee cultivation using 16S rRNA and ITS metabarcoding, analyzing samples from three distinct coffee producing regions, Cauca, Risaralda, and Magdalena.

METHODOLOGY

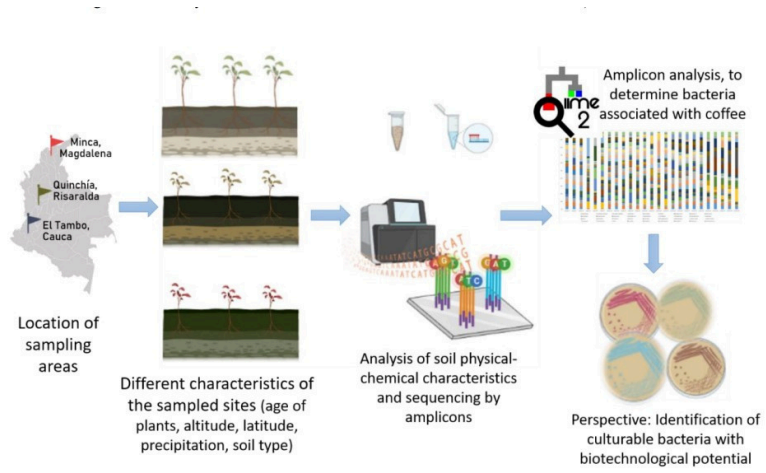


Figure 1. Graphical overview of the methods of this study.

RESULTS

Key genera identified include Janthinobacterium, Bacillus, Actinomadura, and Actinoallomurus, which have potential biotechnological applications such as promoting plant growth and producing antibiotics. Additionally, the study explored soil fungi from six farms in Colombia, revealing diverse fungal communities, particularly Mortierella and Saitozyma, with varying functional roles.

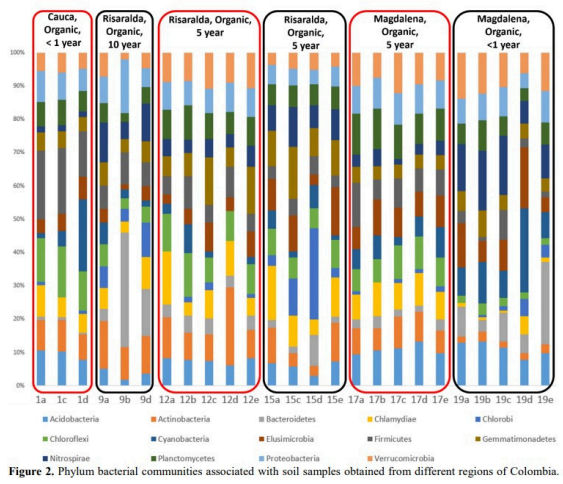


Figure 2. Phylum bacterial communities associated with soil samples obtained from different regions of Colombia.

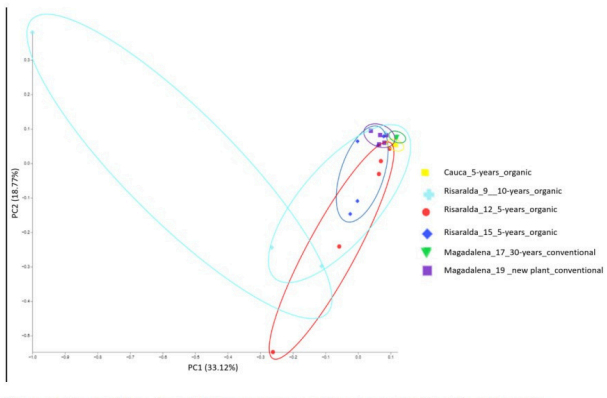


Figure 5. Calculated beta diversity for bacterial communities associated with coffee soil samples.

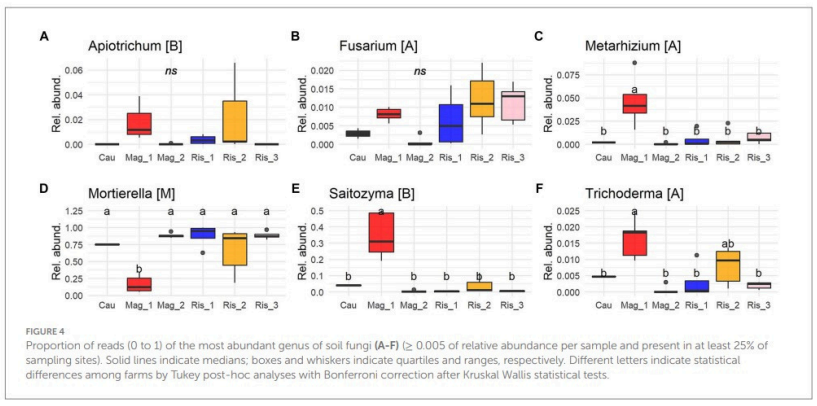
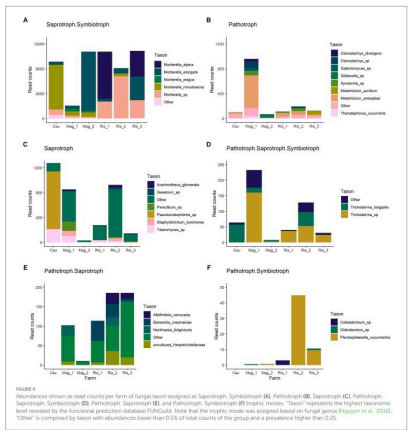


FIGURE 4
Proportion of reads (0 to 1) of the most abundant genus of soil fungi **(A–F)** (≥ 0.005 of relative abundance per sample and present in at least 25% of sampling sites). Solid lines indicate medians; boxes and whiskers indicate quartiles and ranges, respectively. Different letters indicate statistical differences among farms by Tukey post-hoc analyses with Bonferroni correction after Kruskal Wallis statistical tests.



CONCLUSIONS

The research highlights the importance of understanding soil microbiomes for coffee cultivation. The findings underscore the need for further research to examine how different management practices can enhance beneficial fungi and support sustainable coffee production.

ACKNOWLEDGEMENTS

This work was supported by CABANA (Capacity building for bioinformatics in Latin America) grant, on behalf of the Global Challenges Research 408 Fund [BB/P027849/1] and the grant “Capacity building for bioinformatics in Latin America- Network” (CABANA-net), funded by the Chan Zuckerberg Initiative (CZI) under the agreement (2022– 316296)